

Dongqi Yang

Senior iOS Engineer · 10+ yrs · Video/Audio & AI Specialist

github.com/DongQi-Yang | codementor.io/@dongqiyang
Shanghai, China (UTC+8) · overlap with US evenings & EU afternoons

SUMMARY

- 10+ years of iOS development, specialized in **video/audio technology**: FFmpeg, AVFoundation, VideoToolbox, Metal / OpenGL ES, ReplayKit / ScreenCaptureKit; built video-editing and screen-recording engines from scratch.
- Core engineer of a video app family with **500M+ downloads on Google Play alone** (VideoShow 100M+, V Recorder 100M+ — publicly verifiable) and **670k+ App Store ratings** worldwide; led crash governance keeping the live crash rate **under 0.04%**.
- AI in production: LLM API integration (streaming, async generation pipelines), on-device Core ML / TFLite / MLKit, and a 64-capability AI-services SDK; power user of AI-assisted engineering with compile-gated, test-gated pipelines.
- Also ship Flutter (hybrid + platform channels) and led a Flutter → native SwiftUI migration with Clean Architecture.

CORE SKILLS

Swift SwiftUI Objective-C C++ FFmpeg AVFoundation VideoToolbox Metal OpenGL ES
ReplayKit ScreenCaptureKit Core ML TensorFlow Lite Flutter / Dart StoreKit 2 / IAP Firebase
REST / WebSocket SPM / Modularization TDD / XCTest App Store Delivery

EXPERIENCE

Shanghai Yingzhuo Information Technology Senior iOS Engineer 2017.05 – present

- Video Editing Engine (VideoShow, flagship)** — built from scratch: FFmpeg demux/transcode/composition pipeline + VideoToolbox hardware codecs; Metal / OpenGL ES effect rendering; filter-graph (node/pin) core with command-based undo/redo; designer-facing effect-asset protocol so new effects ship **without app releases**. Powers an app with **100M+ Google Play downloads**; crash rate < 0.04% (tools-category average ≈ 0.3%).
- Screen Recorder (iOS + macOS)** — ~10k-line C++/ObjC++ engine inside ReplayKit's Broadcast Upload Extension under the 50MB memory cap: hardware-encode-on-arrival, hand-written **fMP4 muxer** whose moov-first design keeps files playable through crashes; dual Metal/GLES render paths with zero-copy YUV textures; Live Activity (Dynamic Island) cross-process status; macOS version on ScreenCaptureKit + SwiftUI multi-window controls.
- AI Services SDK** — designed a Swift Package exposing **64 AI capabilities** (text-to-image/video, background removal, face swap, voice cloning, music generation, OCR...) behind one interface: unified AWS S3 / Tencent COS multi-cloud upload, cancellable async task lifecycle, per-feature plug-in files. Serves multiple apps and Flutter hosts.
- Flutter → SwiftUI migration (Waazy, AI music)** — led module-by-module migration to Clean Architecture; extracted native network/membership/AI layer serving both stacks; built an AI-assisted migration pipeline (auto page restoration → compile gate → UI parity audit). Also own StoreKit 2 subscriptions & points system.
- Built 8 in-house SPM libraries (networking + DES signing, analytics, points/receipt validation, ads, performance monitor, AI platform) reused across all company apps; trained junior engineers and reviewed technical designs.

Shanghai Samu E-Commerce iOS Team Lead 2015.07 – 2017.03

- Led the iOS team building cross-border e-commerce app "SSAM" from zero: architecture (MVC + CoreData), WeChat/Alipay payments, analytics & crash monitoring, coding standards and release management.

OPEN SOURCE & WAY OF WORKING

- Interview Workshop** (TypeScript/Express — github.com/DongQi-Yang/interview-workshop) — local web tool with a provider-abstraction layer switching between the Claude API and Apple's on-device Foundation Models with automatic fallback; atomic serialized JSON persistence; 58 tests, zero-network suite; built fully TDD with review-gated AI-assisted workflow.
- Every delivery ships with automated tests, a short demo recording, and daily written updates.

EDUCATION & CERTIFICATES

Northeast Petroleum University | B.Eng., Electronic Information Engineering | PMP Certified

2011 – 2015